Diploma in Data Analytics

Logistic regression

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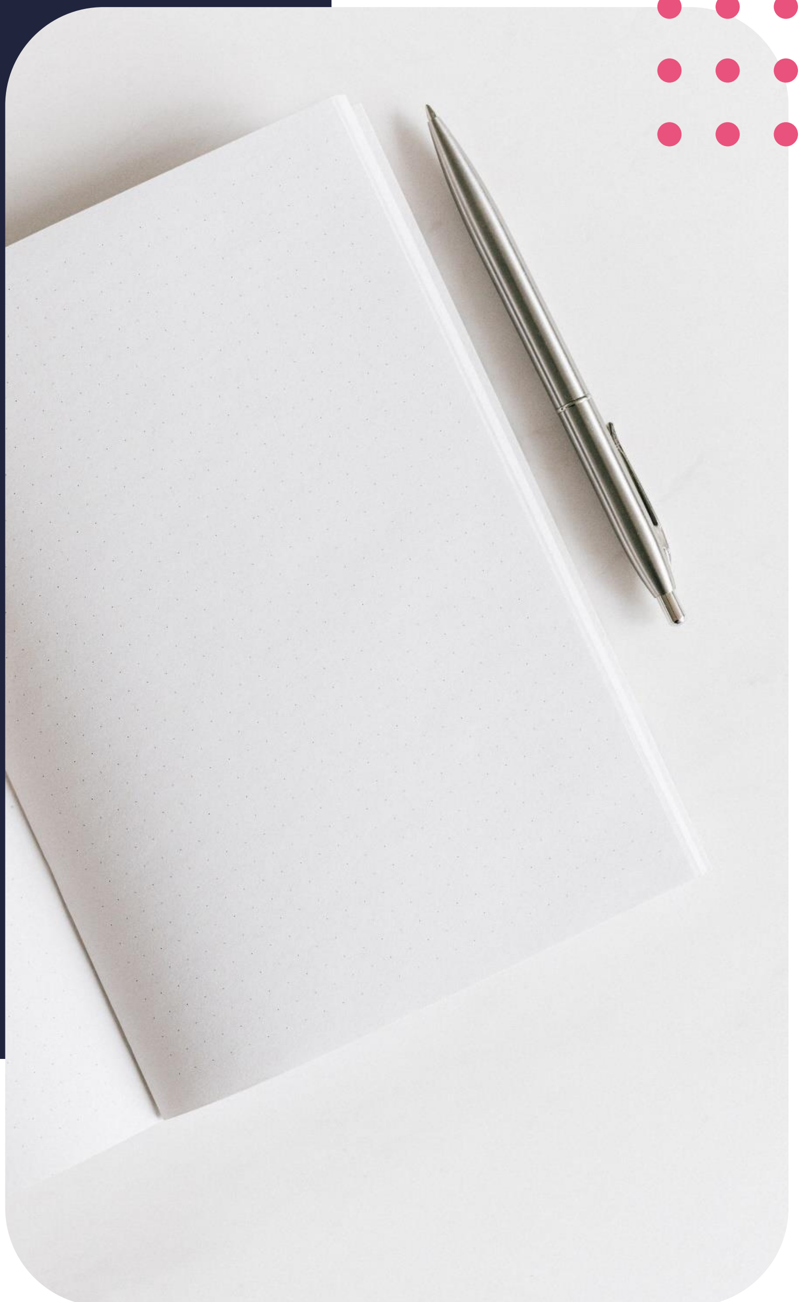


Lesson outcomes

By the end of this lesson, you should be able to:

- Understand when to use linear and logistic regression
- Know when the linear model does not fit the data well

Introduction

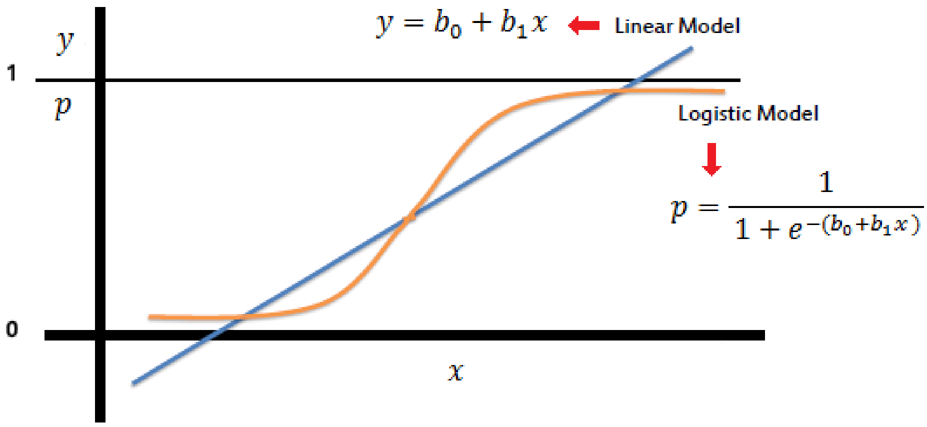
In this lesson, we will elaborate on the introduction to logistic regression from lesson 7. We will better understand when to utilize this model and how to interpret the outcome. Thereafter we will elaborate on the model fit statistics we have briefly touched on in previous lessons, such as the AIC and BIC statistics. We will end the lesson by cementing in all the knowledge we have gained through module 2 with a practical demonstration.

Logistic regression

Recap logistic regression

In the previous lesson, we learnt that Binary logistic regression predicts if something is true or false, did the passengers survive or not? This can also be extended to include various classes of more than 2 events.

Unlike linear regression, logistic regression fits an s-shaped curve to the data that spans from 0 to 1. The curve therefore shows us the probability of the passengers surviving based on the predictor variables we included in the model. But what is an s-shaped curve, you might have been wondering in the previous lesson?



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Logistic regression is similar to linear regression, but the curve is created utilizing the natural logarithmic odds ratio rather than the probability of the target variable. A logistic regression model constructs a logistic curve that is s-shaped with values limited between 0 and 1. And unlike linear regression, the predictor variables do not have to be normally distributed or have equal variances from group to group.

Mathematically, the logistic regression model can be written as:

$$p = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$$

$$\frac{p}{1-p} = \exp(b_0 + b_1 x)$$

$$\ln(\frac{p}{1-p}) = b_0 + b_1 x$$

Through a transformation of this formula, we can write the logit function in terms of an odds ratio.

Multiplication with the natural log on the left-hand side and right-hand side of the equation, will lead to a linear form function of the predictors.

This function is interpreted as follows: the b1 coefficient is the amount the log-odds changes for every one-unit increase or decrease in x.

There are many similarities between the ordinary least-square regression and the logistic regression model. Where ordinary least squares regression is used to estimate the coefficients that best fits the line for the linear regression model, logistic regression in turn uses the maximum likelihood estimation we learnt about at the start of this module to procure the coefficients for the model. After the beginning of this initial function/assignment, the method is repeated until the log likelihood stops converging significantly.

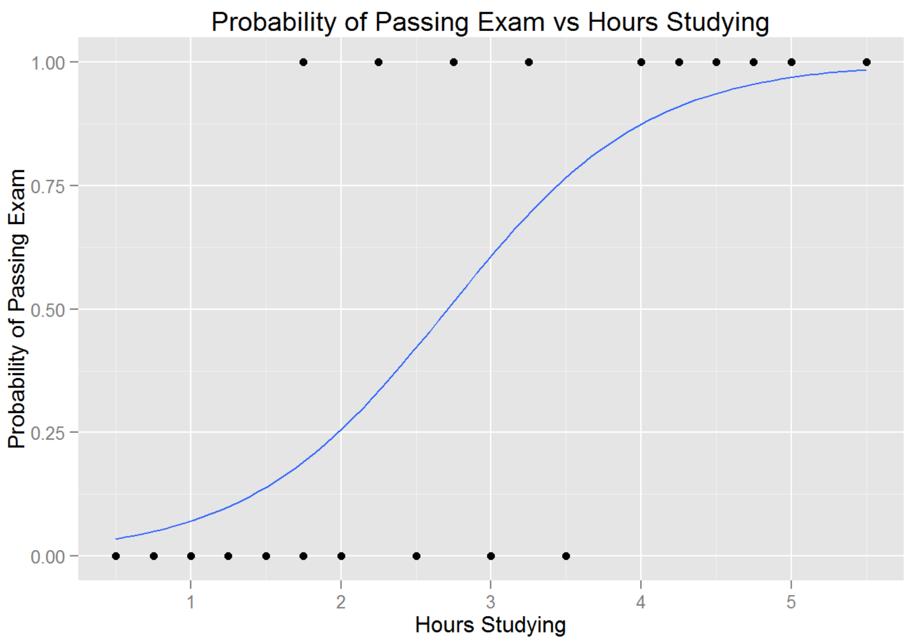
Examples of logistic regression

- The banking industry often makes use of logistic regression models. For example, a bank often has to decide whether a client will default on a loan or not based on a given set of predictors variables. They use historical data about current clients to determine if a potential loan applicant is a good or bad credit risk.
- Another situation where logistic regression can be used for predicting if a tumour is benign or malignant. We could look at factors such as the size of the tumour and the body area that is affected to classify the tumour as malignant or benign.
- Let’s investigate an example of a student passing an exam to better understand logistic regression:

Hours	**Pass**
0.50	0
0.75	0
1.00	0
1.25	0
1.50	0
1.75	0
1.75	1
2.00	0
2.25	1
2.50	0
2.75	1
3.00	0
3.25	1

3.50	0
4.00	1
4.25	1
4.50	1
4.75	1
5.00	1
5.50	1

This table shows whether students passed (1) or failed (0) the examination based on the amount of hours studied



The above plot is created with the ggplot function and a smoother is applied to the curve in order to better understand the trend of the data. The plot fit a logistic regression curve to the given data that shows the probability of the student passing depending on the number of hours they studied for the test.

When fitting a logistic regression model to the data, we see that the output shows the number of hours studying is significantly related to the probability of the student passing the exam. The coefficient for the intercept is -4,078 and the coefficient for the number of hours studying is 1,505.

The probability of the student passing can be mathematically expressed in the following equation:

$$probability\ of\ passing\ exam = \frac{1}{1 + e^{-(-4{,}078 + 1{,}505(hours\ studying))}}$$

The interpretation of this equation is: every extra hour of study, leads to an increased log-odds of passing the exam by 1,505.

Assumptions of logistic regression model

So just as with the linear regression model, the logistic regression model has some key assumptions we need to be aware of when using this model.

- Unlike linear regression, logistic regression does not require any of the fundamental assumptions of the linear model regarding linearity, normality, homoscedasticity (which has to do with the relationship between the error terms) and measurement level. However, binary logistic regression does require the dependent variable to be binary and ordinal logistic regression calls for the dependent variable to be ordinal, meaning that we should be able to tell the position of the number when in a list.
- Secondly, the data points in a logistic regression model should be independent of each other, meaning that if the data contains repeated measurements or matched data, it is not suited for the logistic regression model.
- Like linear regression, logistic regression also calls for little correlations to exist between the independent variables.
- Fourth, logistic regression requires that the independent variables be linearly related to the log odds, meaning that this model assumes linearity of independent variables and log odds.
- Lastly, and this is more of a note than an assumption, the logistic regression model generally needs at least 10 observations per variable class in the model

Model fit statistics

So how do we now test which of these models are the best fit for the data? There are various model fit statistics we can use to determine if the model fits the data well:

- R^2, the coefficient of determine
    - Used to measure the variation in the outcome variable. We also know that as the variables we have in the model increases, the R^2 statistic also improves. Therefore, we need to use other statistics as well to determine the model fit
- Residual standard error or RSE statistic.
    - We know that the residual standard error is a measure of how well the linear regression model fits the data.
- Mallow's Cp
- Akaike information criterion (AIC)
- Bayesian information criterion (BIC),
- Adjusted Akaike information criterion
- Adjusted Bayesian information criterion
- Adjusted R^2

All of these are fit statistics that can be used to determine the goodness of the model fit, in other words, to determine if the given distribution fits the model well.

Additional resources

Train, K., 2003, *Discrete Choice Methods with Simulation*, Cambridge University Press, https://eml.berkeley.edu/books/choice2.html

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